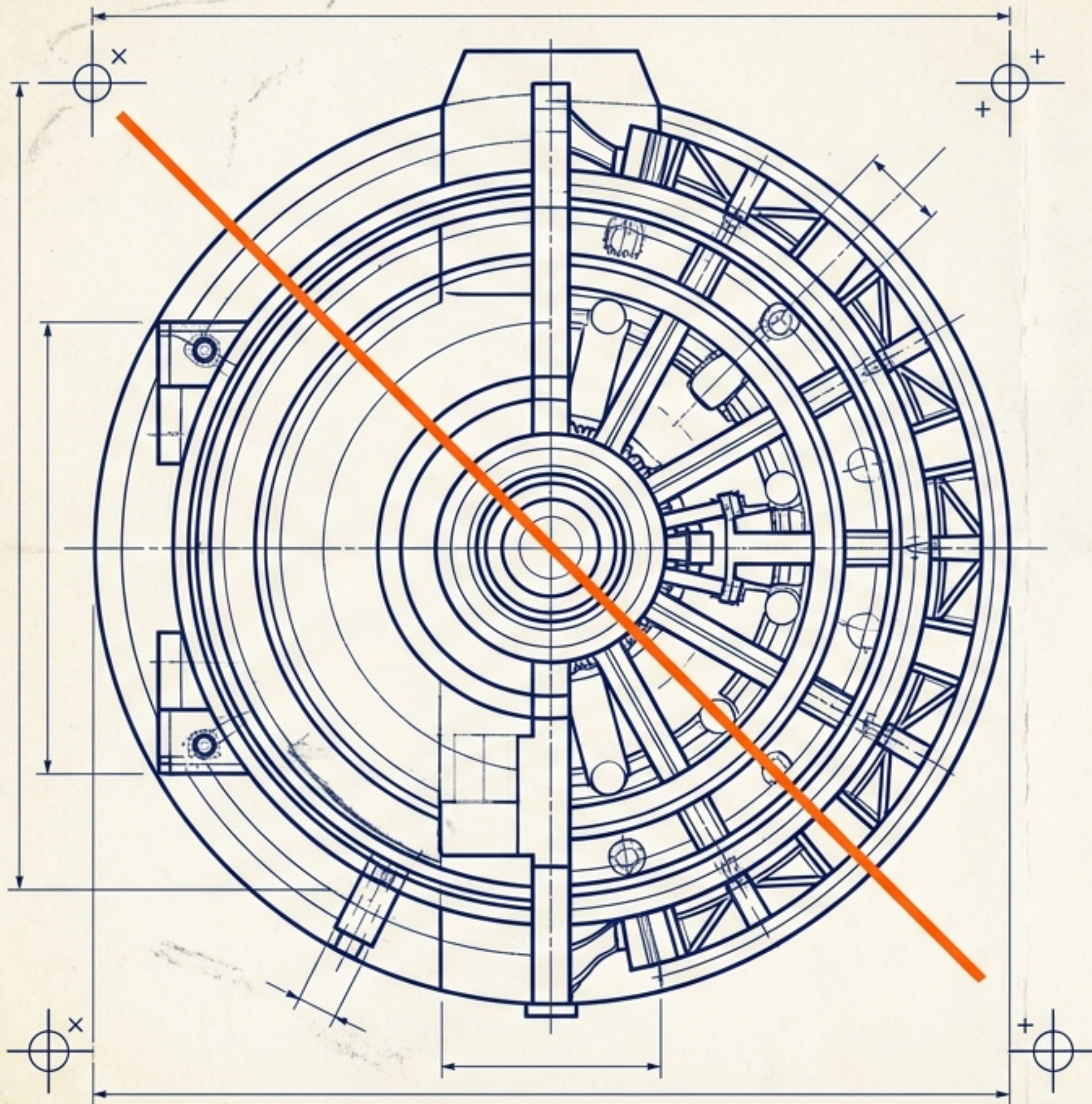
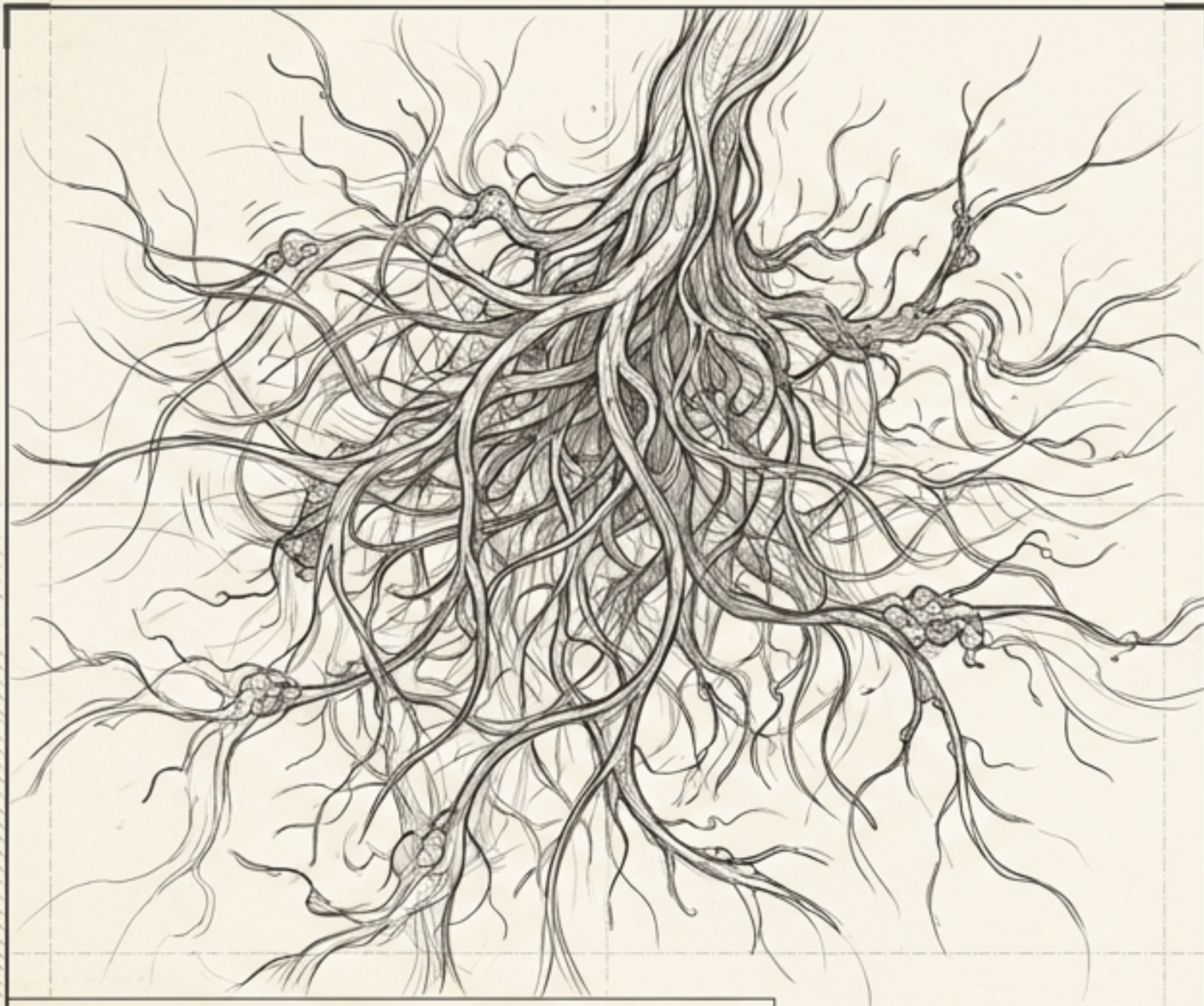




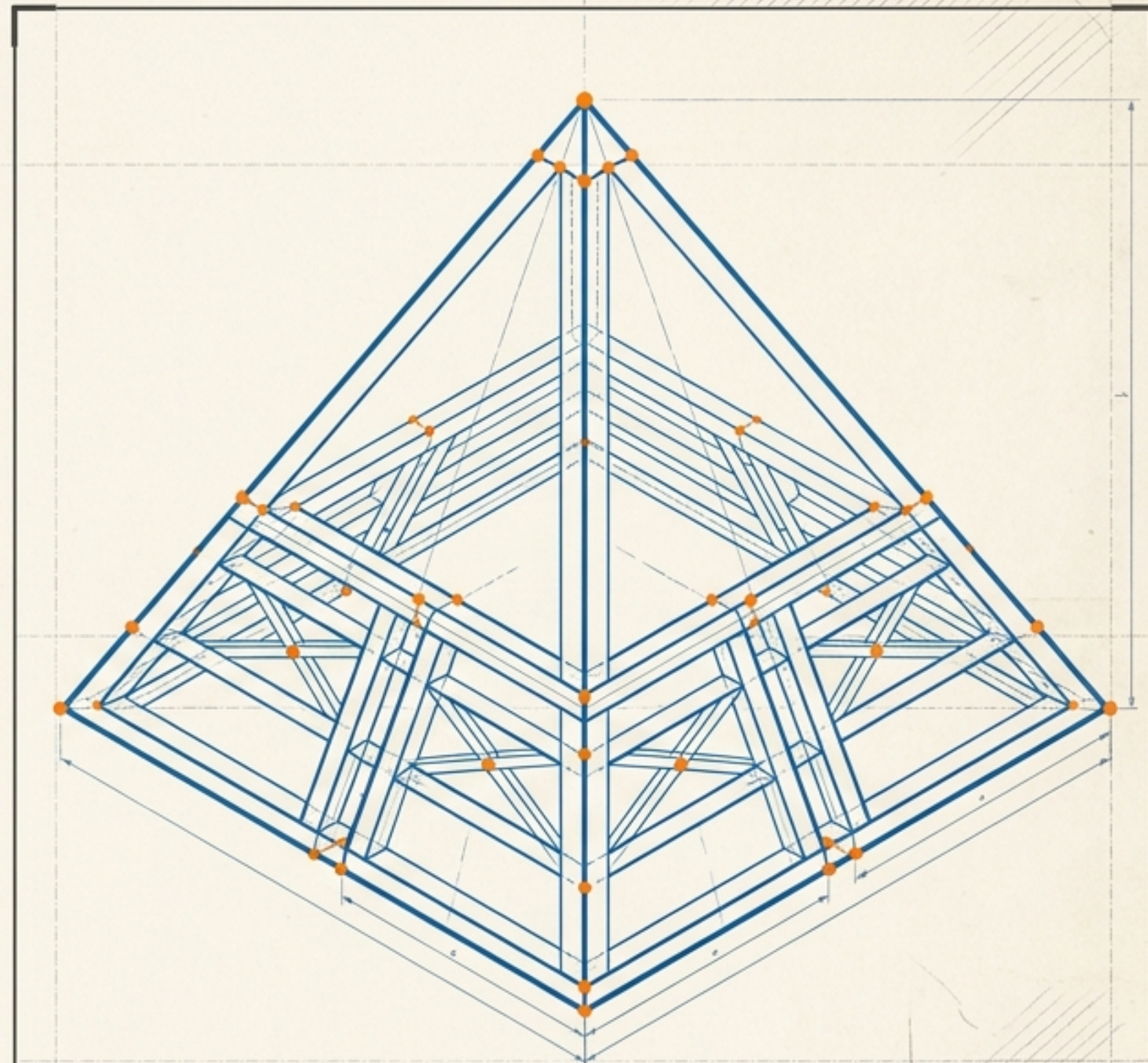
The Austral Language Specification

**Designing strict,
crystalline software
that outlasts its hardware.**

Software is for building
pyramids, not organisms.



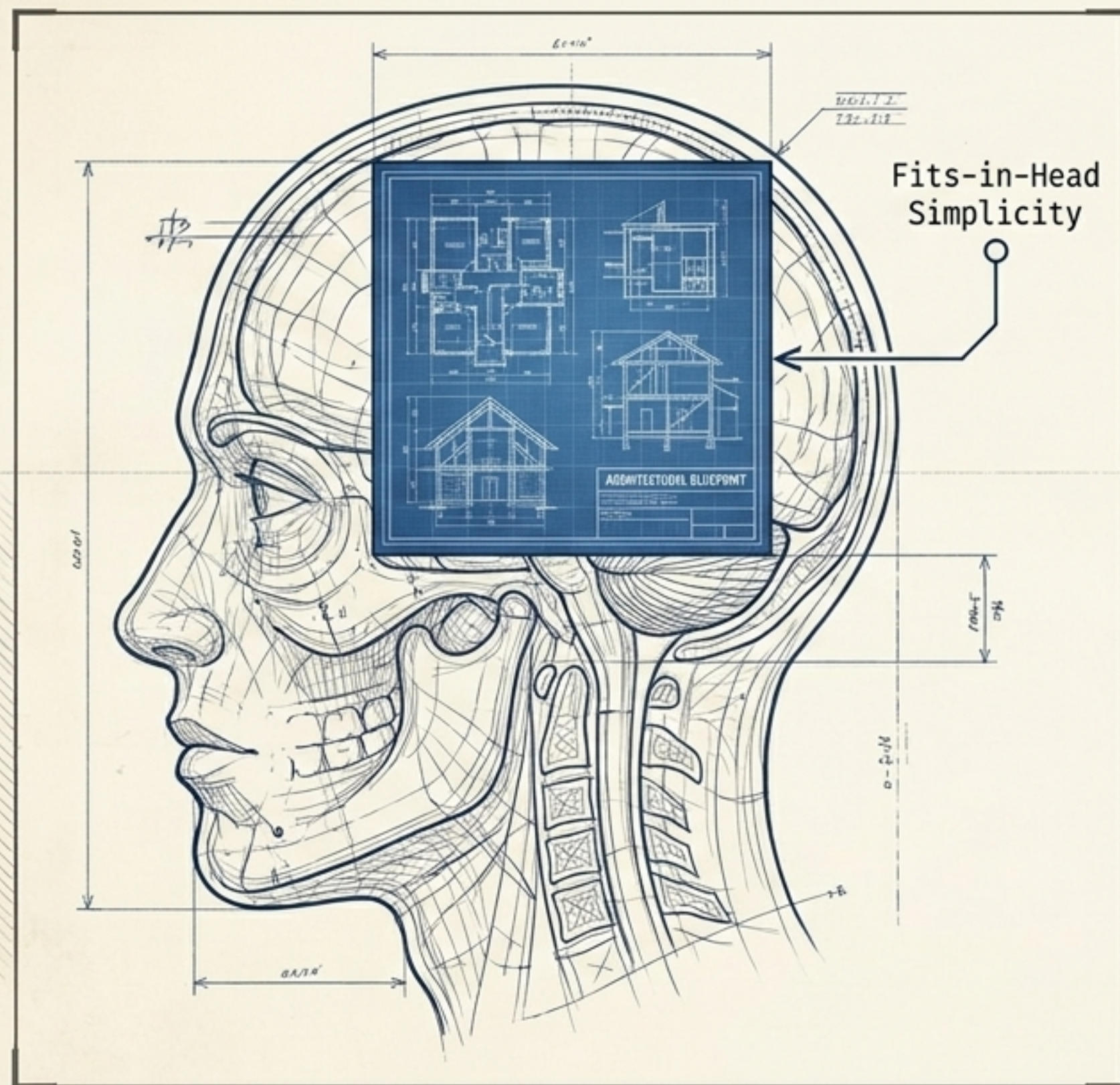
Dynamic, fluctuating myriads (Lisp/Ruby)



Imposing, static structures built by heavy blocks (Austral)

Austral is a language for building **pyramids**. The code is **strict**, **rigid**, **crystalline**, and deliberately **brittle**. Minor changes are prone to breaking the build. This is not a flaw; it is the **ultimate defense mechanism**.

Simplicity is a measure of Kolmogorov complexity, not forgiveness.



A system is simple when it can be described briefly. It does not mean easy to use.

1

Language lawyering must be impossible.



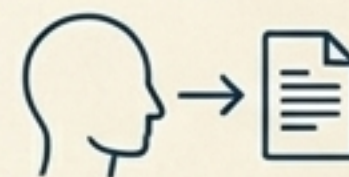
2

Ambiguity is a language failure, not a programmer failure.



3

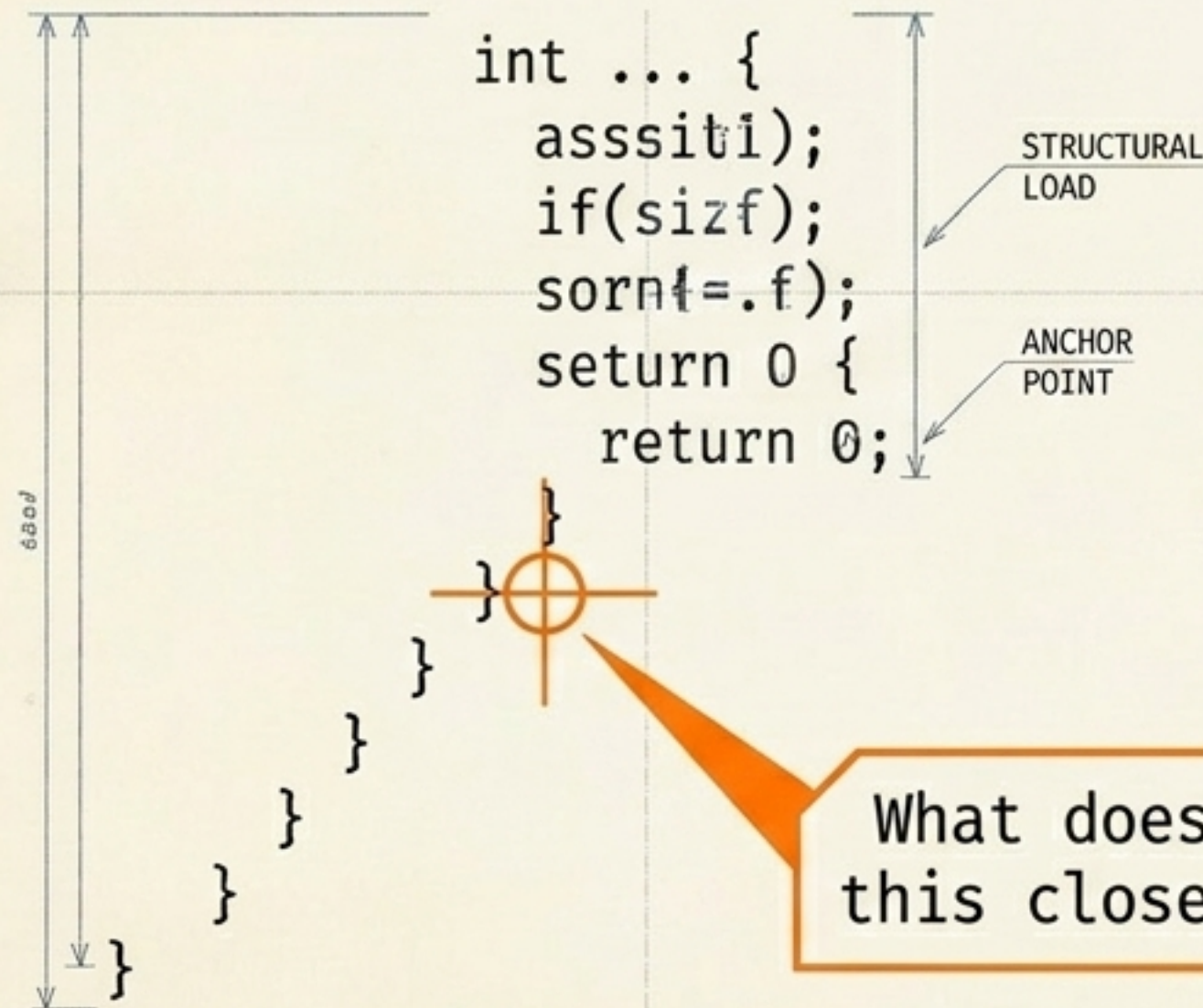
The entire specification must fit in a single person's head.



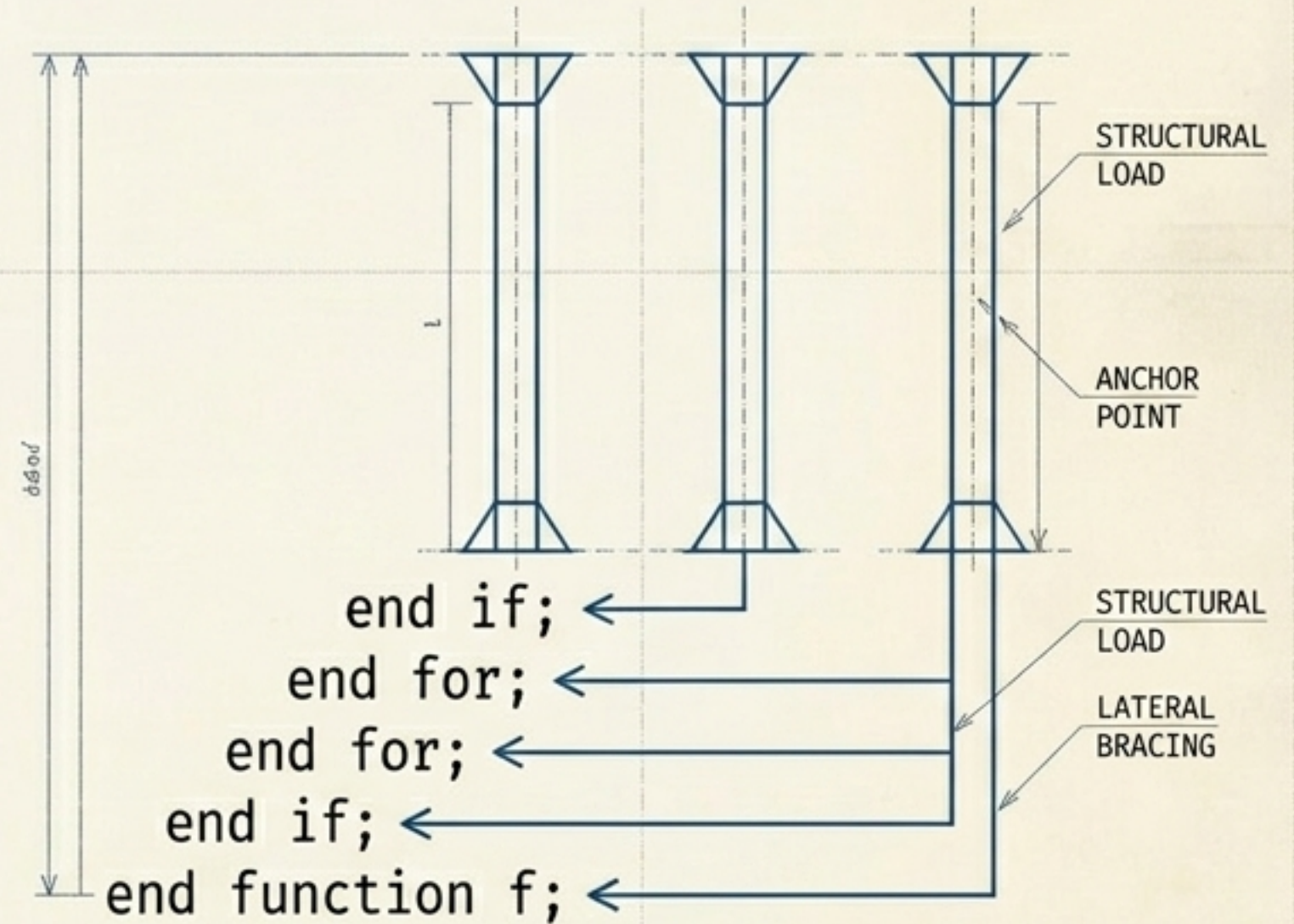
Delimiters act as structural anchors in deep code bases.

Austral favors English-language words over symbols and requires explicit statement termination. Optimize for reading, not typing.

The C++ Sea

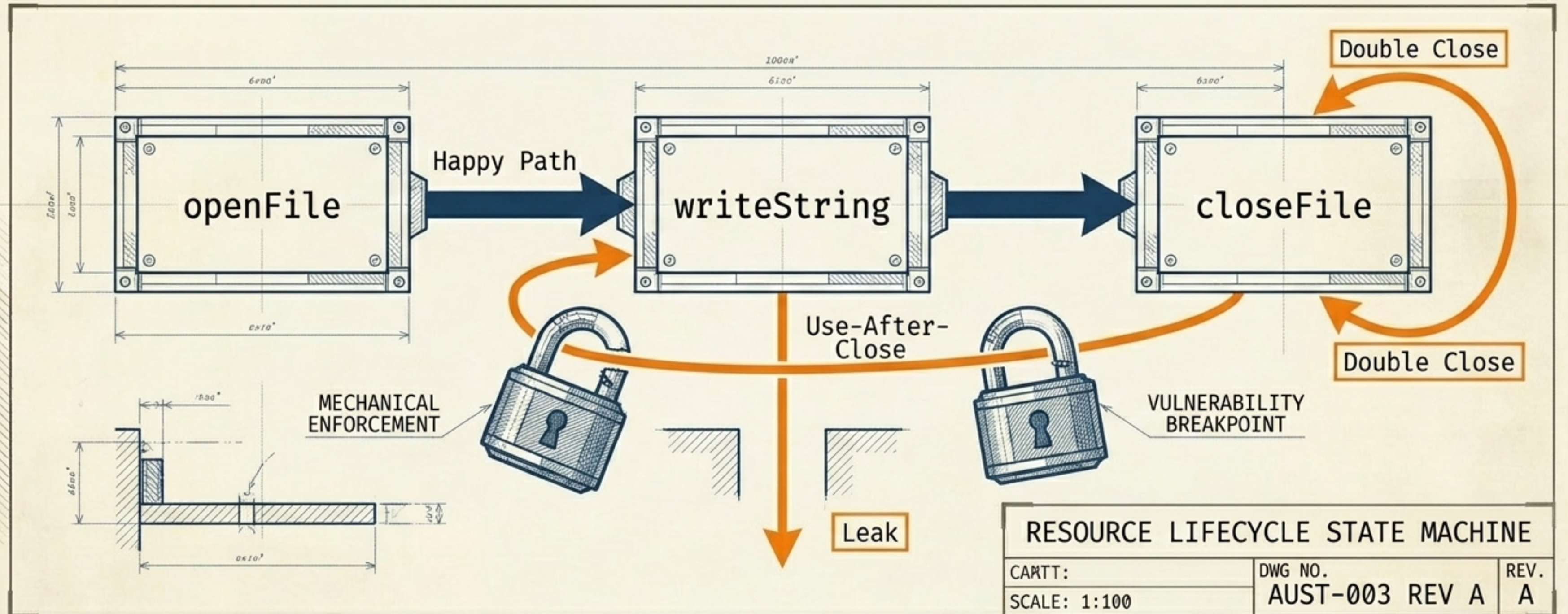


The Austral Architecture



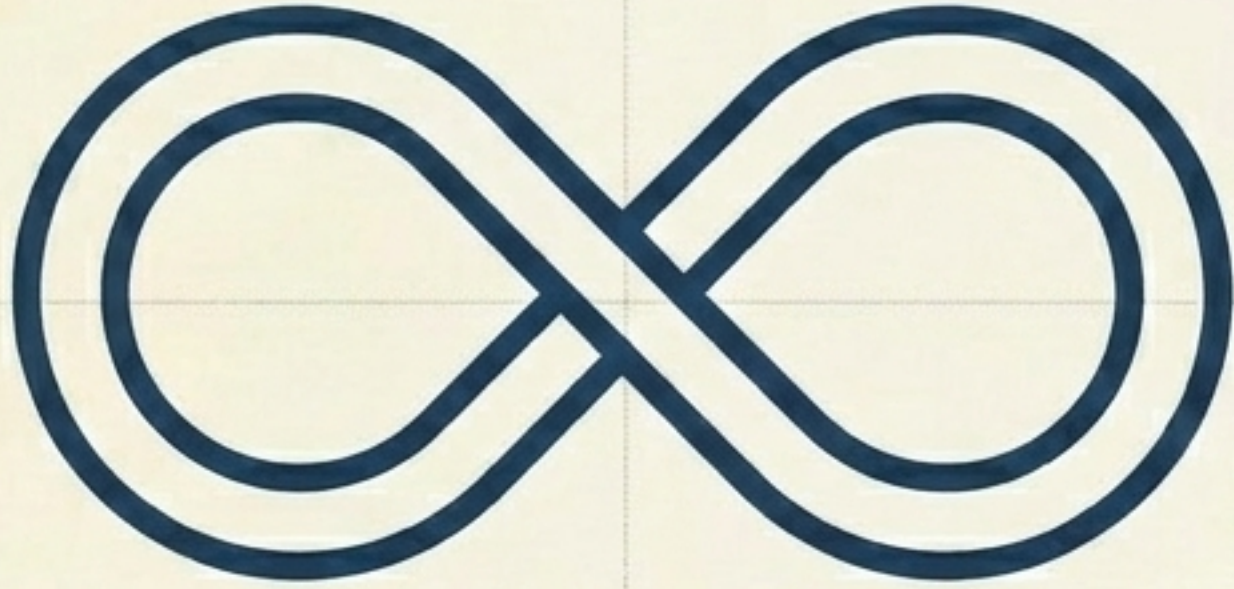
The invisible vulnerabilities of implicit lifecycles.

Traditional APIs rely on programmer discipline to enforce lifecycles. Austral enforces the graph mechanically at compile time.



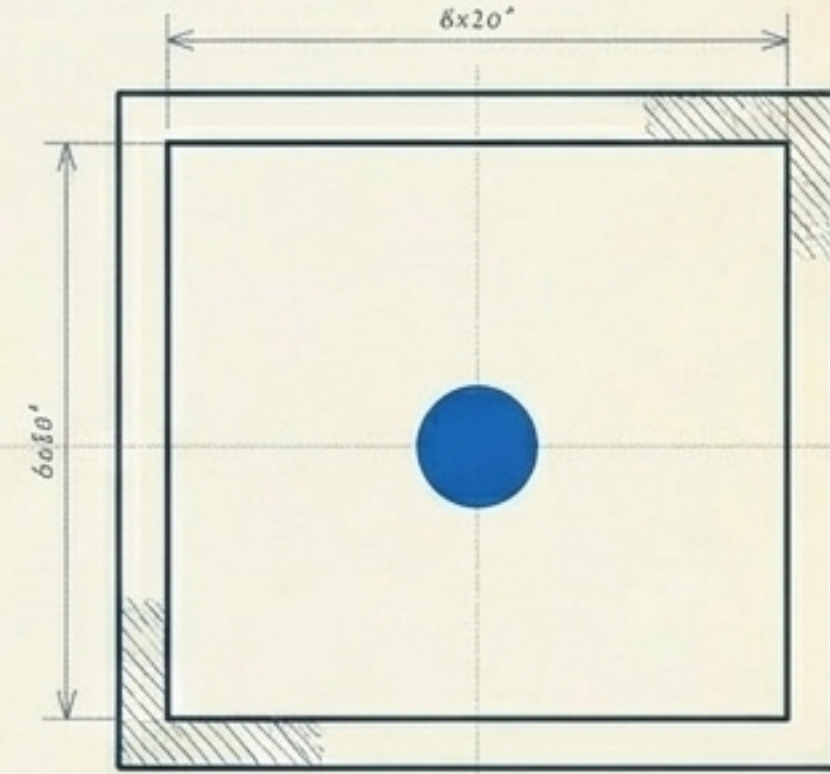
Two universes governing the physics of data.

The Free Universe (Free)



- Infinite cloning allowed
- Silent discarding permitted
- Data: Booleans, Ints, Floats

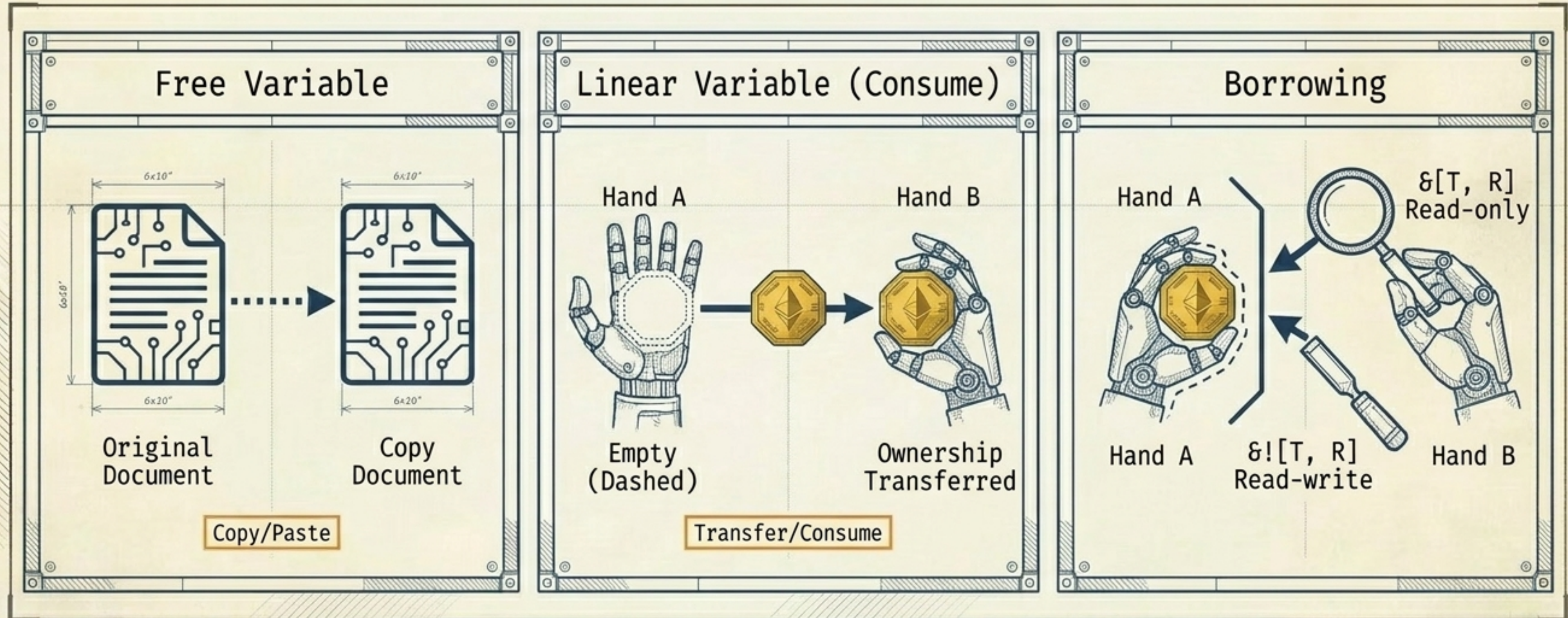
The Linear Universe (Linear)



- The "Use Exactly Once" physics
- Cannot be duplicated
- Cannot be silently dropped
- Data: Resources, Pointers, Capabilities

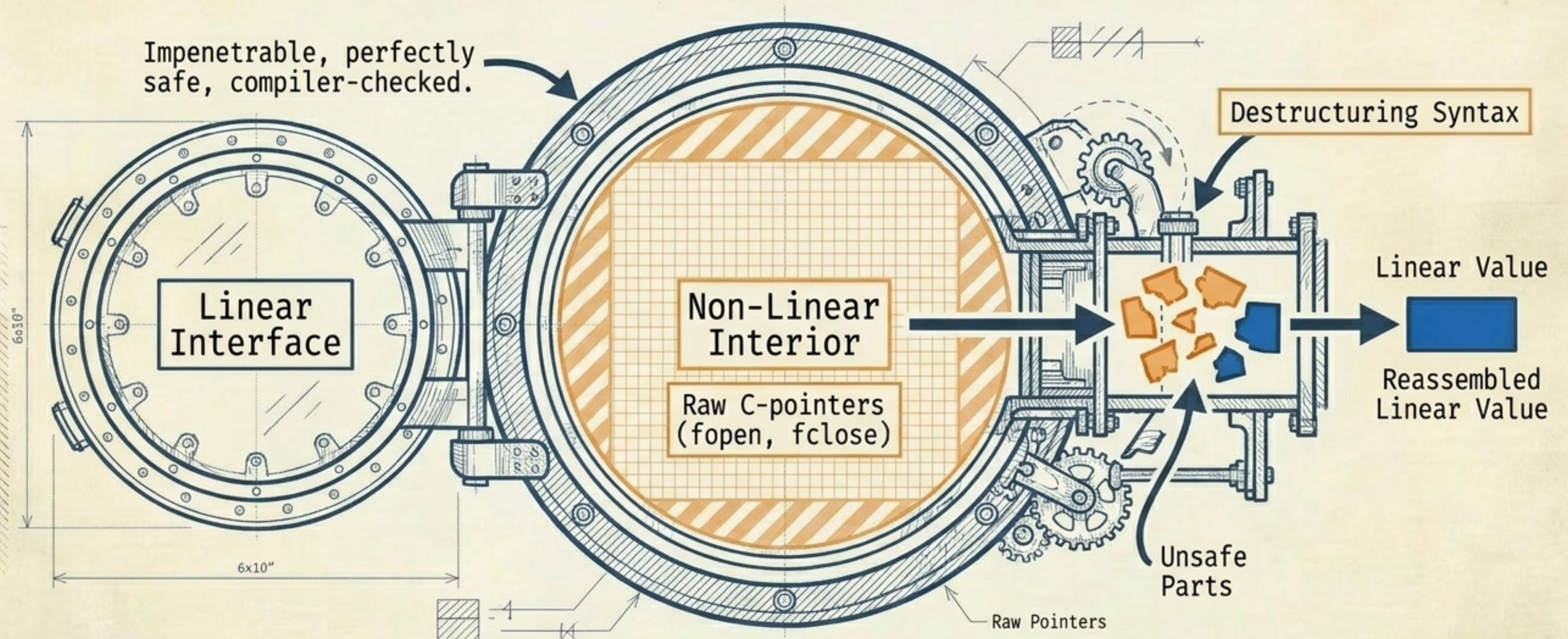
Linear variables are physical objects, not digital copies.

Borrowing gracefully degrades permissions, allowing functions to read or mutate a linear value without taking ownership.

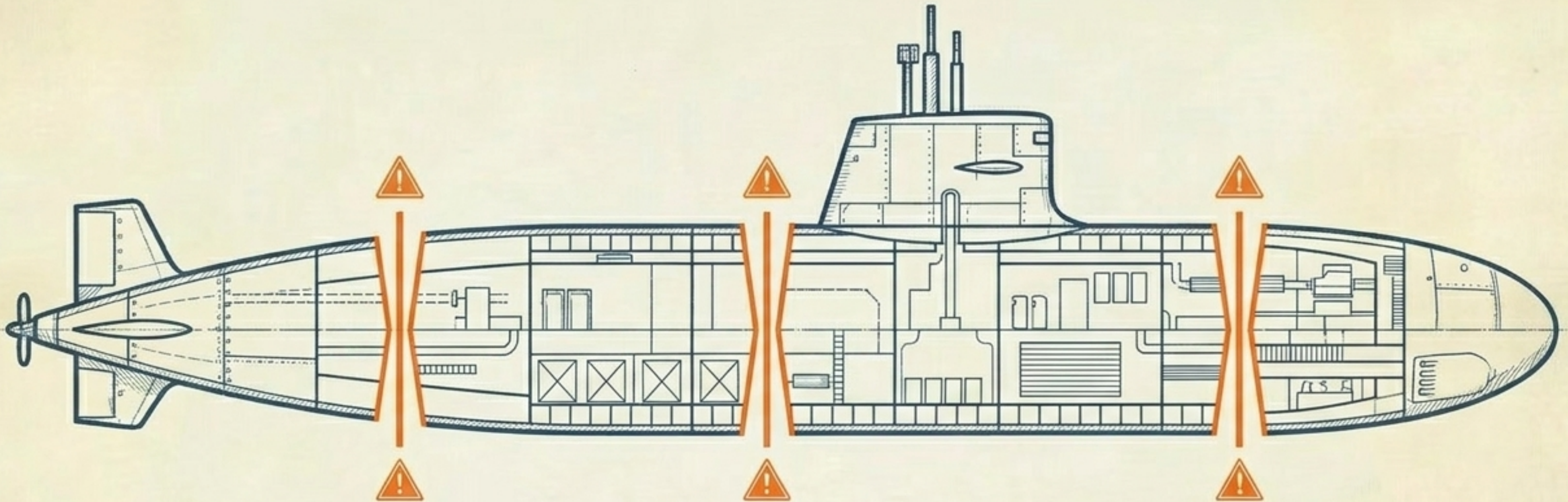


The Trust Boundary: Non-linear interior, linear exterior.

Carefully vetted unsafe FFI code operates inside. The client only sees a strict linear interface that can never be misused.



On contract violation, scuttle the ship without delay.



When a program enters an invalid state, you must assume an adversary is in the system. Recovery efforts are attack vectors. Austral's response to abstract machine corruption is brutal and instantaneous: Terminate Program. No stack unwinding. No cleanup code.

Why exceptions fail the simplicity and security mandate.

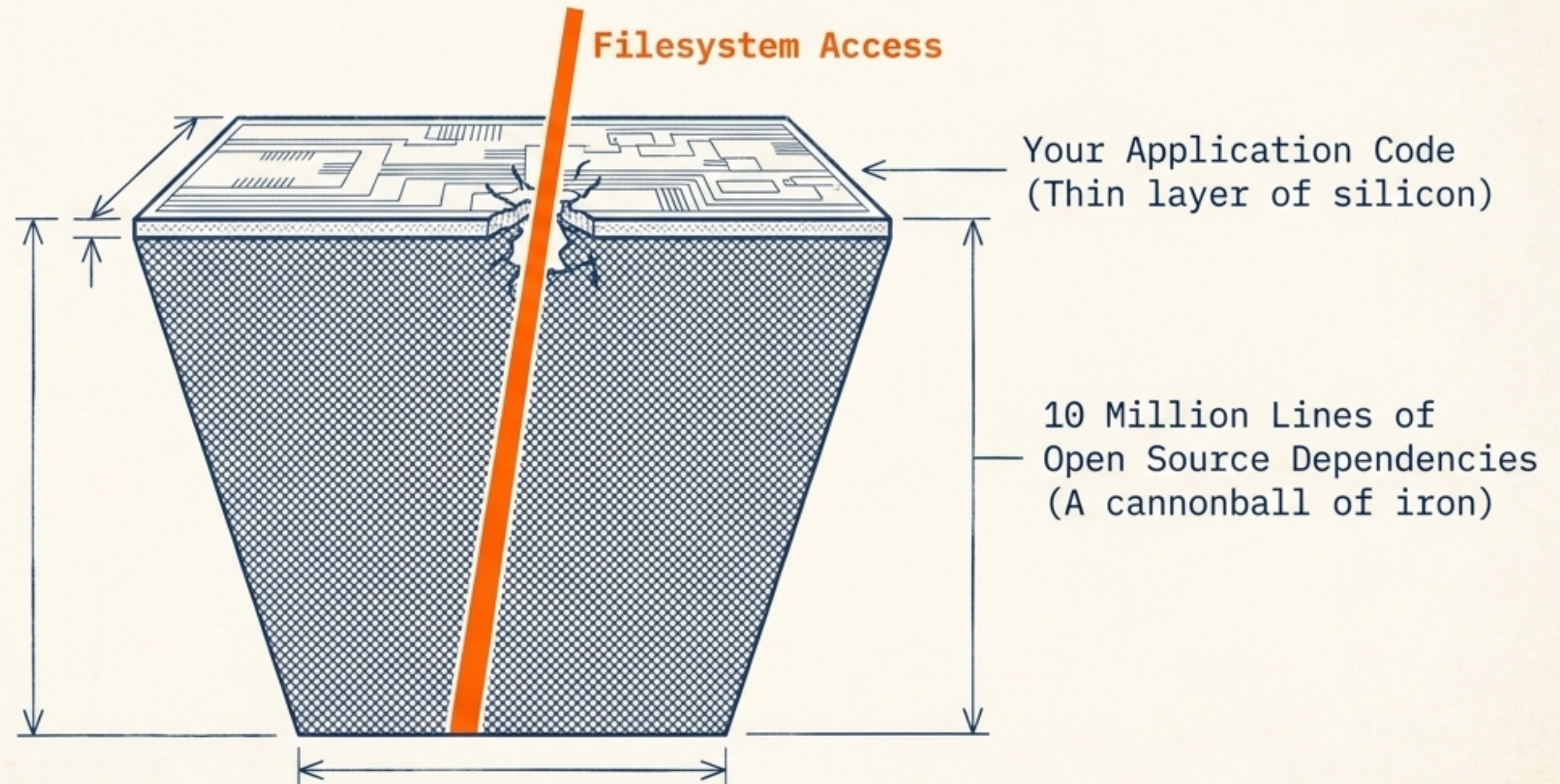
Linear types guarantee resource safety on the happy path. Exceptions introduce unpredictable control flow and bypass the type system.

DIAGNOSTIC MATRIX - REV 1.4

	Terminate Program (Austral)	Terminate Thread	Raise Exception (Unwind)
Semantic Complexity	None	Moderate	Extreme (Hidden control flow everywhere)
Attack Surface	Zero (Instant death)	Low	High (Destructors execute mid-attack)
The "Double Throw" Problem	Immune	Immune	Pathological (Throwing during a destructor crash)

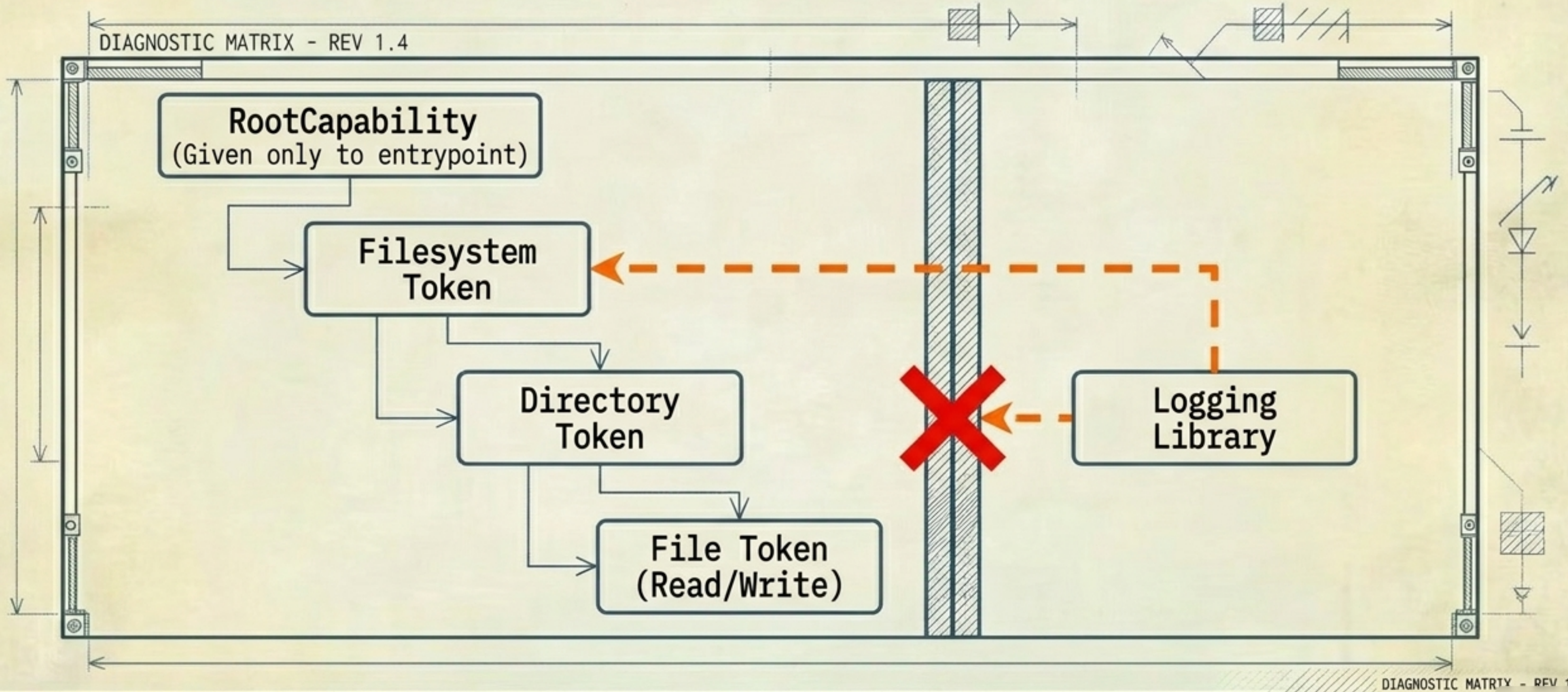
The danger of uniform root permissions

Code is overwhelmingly permissionless. An innocent transitively imported library can silently read `/etc/passwd` and ship it over the network.



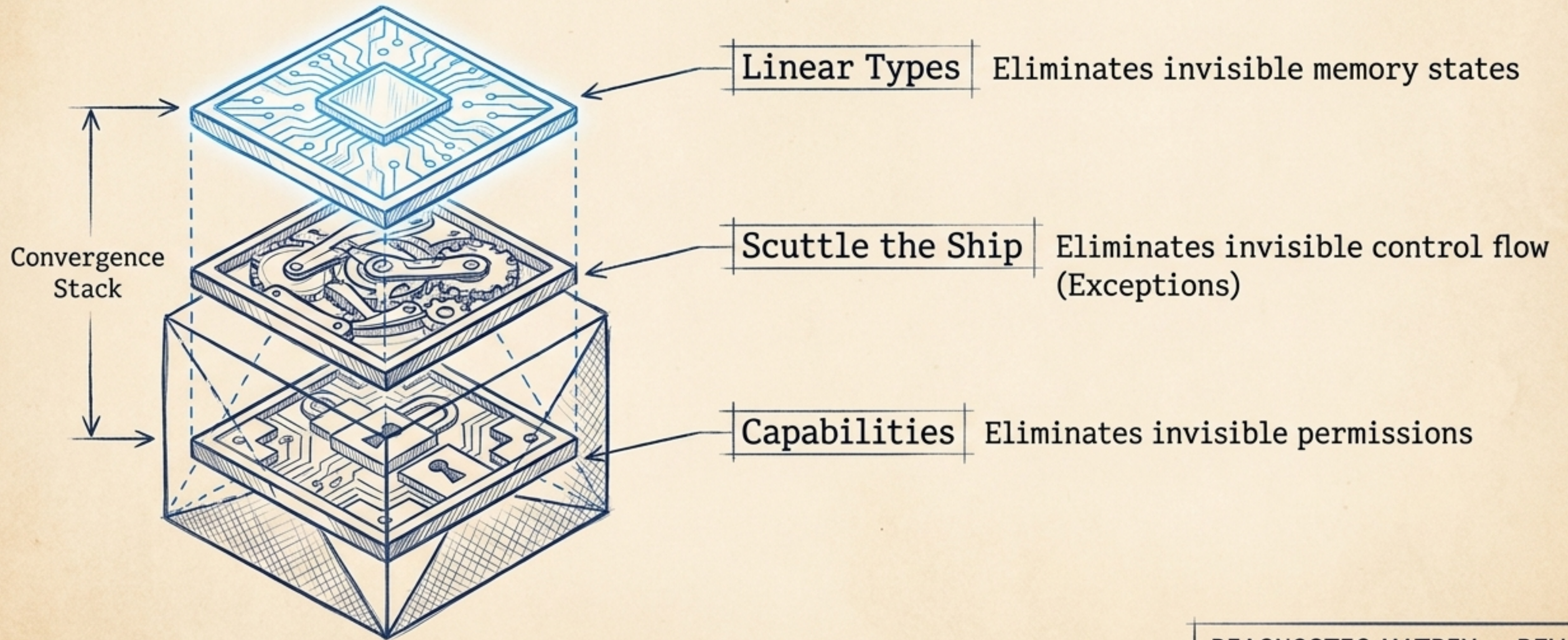
Capabilities mechanically eliminate supply-chain attacks.

A capability is an unforgeable linear token. If a library's function signature does not demand a capability, it has zero access to the outside world.

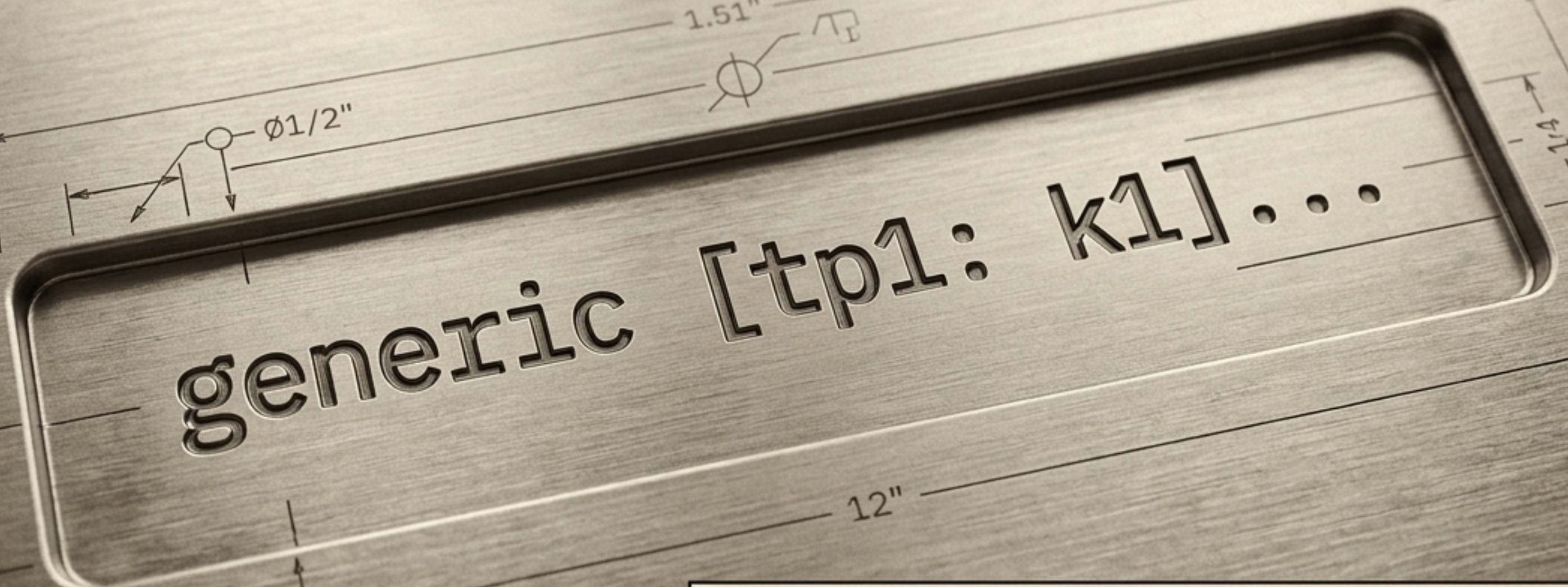


The compiler ruthlessly eliminates invisible state.

“Fits-in-head” simplicity isn’t about having a simple syntax. It is the result of the system absorbing the 90% of cognitive load usually spent tracking lifecycles, unwinding paths, and security audits.



An automobile requires maintenance;
a mathematical expression does not.



A technical drawing of a rectangular metal plate with rounded corners. The plate is oriented diagonally. It features several dimensions: a total length of 12 inches, a width of 8 inches, and a corner radius of 1.51 inches. There are two circular holes, one with a diameter of 1/2 inch and another with a diameter of 1.51 inches. The text "generic [tp1: k1]..." is engraved on the plate. The drawing includes various dimension lines, arrows, and a section line symbol.

generic [tp1: k1]...

Bitrot is not inevitable. By prioritizing absolute stability, restraining programmer power, and mechanically aiding correctness, Austral is designed to compile, run, and remain secure—without changes—decades into the future.